

YOUSSEF YASSER ELSAYED HAMED

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PROFILE

Third-year CS and AI student at Cairo University specializing in Machine Learning, Deep Learning, and Computer Vision, with hands-on experience building end-to-end ML systems for real-world applications. Strong in problem-solving, model optimization, and developing reliable AI solutions that are practical, scalable, and production-ready.

EDUCATION

Cairo University

Bachelor's in Computer Science and Artificial Intelligence

Giza, Egypt

Sep. 2023 – Sep. 2027

EXPERIENCE

AI Engineer Intern

Novus

April 2026 – Present

On-site, New Cairo, Egypt

- **Behavioral AI Theft Detection System** using deep learning and computer vision to detect suspicious human-object interactions in retail-like environments.
- **Deploying AI solutions** within a professional engineering environment, ensuring robust model performance, maintainable code quality, and continuous system improvements.
- **Improved system reliability** through model evaluation, pipeline optimization, and iterative testing.

Machine Learning Intern

Code Alpha

April 2026

Remote

- **Handwritten Character Recognition:** Built an image classification system across **47** alphanumeric classes using the EMNIST dataset using 131,000+ training images, achieving **94%** accuracy.
- **Disease Prediction Model:** Developed a binary diagnostic classifier for tumor malignancy using the Breast Cancer Wisconsin dataset using 560+ clinical instances, achieving **96%** accuracy.

Competitive Programmer

Egyptian Collegiate Programming Contest (ECPC)

July 2024

Alexandria, Egypt

- Engineered optimized algorithmic solutions in C++ under strict time constraints, applying advanced data structures to tackle complex computational problems.
- Collaborated in a high-pressure team environment to prioritize problem selection, rapidly debug logical errors, and maximize competitive performance.

PROJECTS

Adaptive Pose & Action Intelligence System [GitHub] | *Python, PyTorch, YOLOv8, OpenCV, SQLite* May 2026

- **Multi-Level Human Analysis Pipeline:** Engineered a real-time computer vision system simultaneously tracking **527** anatomical points across 3 analysis levels: **17** body keypoints via YOLOv8-Pose, hand state estimation, and **68** facial landmarks via YuNet + LBF — achieving **55 FPS**.
- **Action & Posture Intelligence:** Extracted **8** joint angles per frame to compute real-time posture scores (0–100) and classify **6** body actions using a hybrid rule-based + PyTorch **LSTM** pipeline operating on 30-frame temporal sequences.
- **3-Layer Memory System:** Implemented a persistent behavioral memory using short-term RAM deque, unsupervised pattern learning across sessions, and a **SQLite** long-term store — enabling cross-session posture trend tracking and anomaly flagging with threaded face analysis for optimal throughput.

Behavioral AI Theft Detection System [GitHub] | *Python, PyTorch, YOLOv11, SAHI, OpenCV* April 2026

- **Real-Time Vision Pipeline:** Engineered a computer vision pipeline using **YOLOv11x** for high-fidelity pose estimation and instance segmentation, actively monitoring **30** high-value target classes.
- **Robust Detection & Tracking:** Integrated **SAHI** (1000×1000 tiling) to maximize small-object recall, with **ByteTrack** and a **ResNet18**-based ReID manager for persistent multi-target tracking across occlusions.
- **Behavioral Classification:** Extracted **17** dynamic spatial features (grab motion, joint angles) to train a custom PyTorch **LSTM** network with an attention mechanism for classifying suspicious temporal movements.

Plant Disease Recognition System [Kaggle] | *Python, TensorFlow/Keras, CNN, OpenCV* March 2026

- **Automated Diagnosis:** Developed a deep learning system classifying **38** distinct plant disease categories using the PlantVillage dataset using 54,000+ images, achieving **99%** validation accuracy.

- **Model Architecture & Training:** Designed and optimized a **Convolutional Neural Network (CNN)** pipeline for high-resolution image classification, efficiently analyzing over **54,000** images across various species.
- **Comprehensive Evaluation:** Validated model performance using Confusion Matrices, Precision-Recall metrics, and per-class accuracy analysis across all 38 disease categories.

Traffic Sign Image Classifier [GitHub] | *Python, PCA, KNN, Naive Bayes, MobileNet V2* Feb. – March 2026

- **Traffic Sign Classification:** Developed a traffic sign classification system for **43** classes using 39,000+ images, achieving **98%** accuracy in road sign categories using the GTSRB dataset.
- **Dimensionality Reduction:** Applied PCA to reduce feature space while retaining **95%** of variance, improving training efficiency by **60%**.
- **Model Comparison:** Benchmarked classical ML baselines (KNN, Naive Bayes) against a fine-tuned **MobileNet V2** CNN, boosting final accuracy to **98%**.

Credit Risk Prediction System [Kaggle] | *Python, Scikit-Learn, Random Forest, Pandas* Dec. 2025

- **Risk Assessment:** Developed a binary classifier to predict loan default probability using historical credit data (1,000+ records), achieving **91%** accuracy in identifying high-risk applicants.
- **Feature Engineering:** Performed extensive preprocessing including missing value imputation, categorical encoding, and feature engineering to enhance model stability.
- **Model Optimization:** Compared **Random Forest** and **Logistic Regression** classifiers; applied Grid Search CV for hyperparameter tuning, maximizing F1-score and ROC-AUC performance.

CERTIFICATES & AWARDS

Deep Learning Specialization <i>DeepLearning.AI (Andrew Ng) – Coursera</i>	Feb. 2026
Intermediate Machine Learning <i>Kaggle</i>	March 2026
Intro to Deep Learning <i>NVIDIA</i>	Dec. 2025
Egyptian Collegiate Programming Contest (ECPC) Participant	July 2024

STUDENT ACTIVITIES

Minders Student Activity <i>AI Workshop Participant</i>	Aug. 2025
• Participated in hands-on sessions covering the machine learning lifecycle, dataset management, and model development using Jupyter Notebooks for interactive prototyping and visualization.	

TECHNICAL SKILLS

Programming & Tools: Python, C++, Git, GitHub, Jupyter Notebook
Machine Learning & AI: Deep Learning, Computer Vision, Predictive Modeling, CNNs, Object Detection, Pose Estimation, Instance Segmentation, Random Forest, KNN, Naive Bayes, Logistic Regression , LSTM
Data Science & Analysis: Exploratory Data Analysis (EDA), Feature Engineering, Data Cleaning, Data Visualization, Statistical Analysis, Model Evaluation (ROC-AUC, Precision-Recall, Confusion Matrix)
Libraries & Frameworks: PyTorch, TensorFlow/Keras, Scikit-Learn, OpenCV, YOLO (v11), MobileNet V2, ResNet, Pandas, NumPy, Matplotlib
Software Engineering: Data Structures & Algorithms, Competitive Programming, System Optimization
Soft Skills: Problem-Solving, Critical Thinking, Team Collaboration, Time Management, Adaptability, Professional Communication

LANGUAGES

Arabic: Native or bilingual proficiency
English: Professional working proficiency